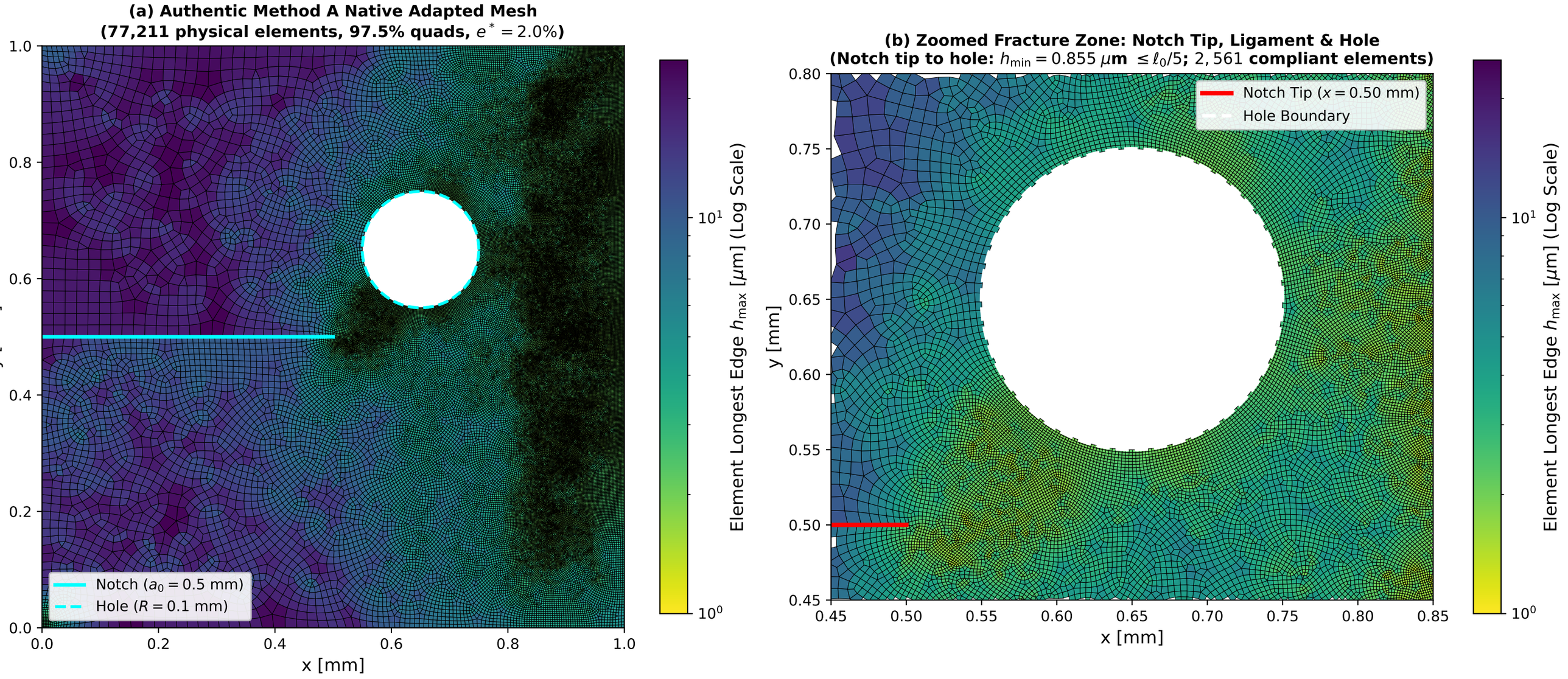

(a) Authentic Method A Native Adapted Mesh
(77,211 physical elements, 97.5% quads, $e^* = 2.0\%$)
Notch ($a_0 = 0.5$ mm)
Hole ($R = 0.1$ mm)
x [mm]
y [mm]
Element Longest Edge h_{max} [μm] (Log Scale)
(b) Zoomed Fracture Zone: Notch Tip, Ligament & Hole
(Notch tip to hole: $h_{min} = 0.855$ μm $\leq \ell_0/5$; 2,561 compliant elements)
Notch Tip ($x = 0.50$ mm)
Hole Boundary
x [mm]
y [mm]
Element Longest Edge h_{max} [μm] (Log Scale)